

Lab: Using Linux for Encryption

Jack Irvin

Challenge Question 1: How does symmetric encryption differ from asymmetric encryption?

- Symmetric encryption uses uniform and consistent encryption across data by having the same public and private key, while asymmetric encryption uses non-uniform encryption by using a public key to encrypt and a private key to decrypt.

Challenge Question 2: If an attacker gains access to this file before encryption, what security risks arise?

- If an attacker gains access to a file before encryption it can be disastrous. They will know the original text of the file as well as all sensitive information contained inside of the file. It will also be possible for them to track down people named inside of the file or change the contents to send a different message. This can be prevented with auditing in place and digital certificates in place.

Challenge Question 3: What does the salt option do in encryption?

- The salt option adds a randomness factor to the contents of a hash being encrypted. It “salts” the file with a pregenerated value to add randomness and make identical inputs have different outputs.

Challenge Question 4: What happens if you use the wrong key for decryption?

- If you use the wrong key for decryption you will not be able to decrypt the file or will receive gibberish. The decryption attempts on the file may also be logged.

Challenge Question 5: Compare AES-256-CBC and Blowfish. Which is more secure and why?

- AES-256-CBC is more secure than Blowfish because it has more resistance to birthday attacks and has a higher bit key. On the other hand, Blowfish has a lower bit key and is more insecure, but it works faster on older systems.

Challenge Question 6: Why is DES considered insecure?

- DES is considered insecure because of its extremely short bit key compared to modern encryption methods, making it easy to be cracked and have attacks launched on it.

Challenge Questions 7

1. What is the difference between brute-force and dictionary attacks?

- The difference between a brute-force attack and a dictionary attack, is that a dictionary attack uses a preconfigured list of words to password guess, while brute-forcing tries every possible combination.

2. Why does DES encryption fail against modern cracking tools?

- DES encryption fails against modern cracking tools because of the short length of its block size. It stands no chance against much faster and stronger password cracking methods that didn't exist when it was first created.

3. How does key length affect the time required for cracking?

- The key length of a password makes the time required for cracking increase exponentially. Longer passwords are much more resilient to password cracking.